

Philosophical Issues, 21, *The Epistemology of Perception*, 2011

TRACKING REPRESENTATIONALISM AND THE PAINFULNESS OF PAIN

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1. Introduction

When you look at the red mug on the desk in front of you, you have an experience with a distinctive phenomenal character. In virtue of what does your experience have this phenomenal character? A popular answer to this question is that given by representationalism, the thesis (at a minimum) that the phenomenal character of an experience supervenes on its representational content.¹ According to representationalism, your experience of the red mug on your desk has its distinctive phenomenal character in virtue of the way it represents the world as being—i.e., in virtue of its representational content.

For many philosophers, representationalism is attractive in part because it seems to lend itself well to the naturalization of phenomenal consciousness. If we can explain the phenomenal character of an experience in terms of its representational content, then by giving a naturalistic answer to the question—in virtue of what does an experience have its content?—we will have gone a long way toward giving a naturalistic explanation of the phenomenal character of experience. The naturalist's answer to this question will have two components. First, she must identify experiences with some naturalistically acceptable entity. Typically, naturalistic accounts of experience will identify token experiences with token neural states.² Second, she must say that these entities have their content in virtue of possessing some natural property. Naturalistic representationalists have tended to favor a “tracking” theory of intentionality, which, very roughly, says that a state represents a property or state of affairs in virtue of causally co-varying with that property or state of affairs under typical or optimal conditions. Different versions of the tracking theory of intentionality will differ in their details, but

for the sake of definiteness, we will take the tracking theory of intentionality to be the following thesis:

- Tracking Theory of Intentionality (TTI): Tokens of a state S in an individual x represent that p in virtue of the fact that: under optimal conditions, x tokens S iff p , and because p .

Suppose your current experience has the content that p . If your experience is identical to your current neural state, then it is your current neural state that has the content that p . In virtue of what does your current neural state have this content? According to TTI, your current neural state has the content that p in virtue of being a token of some (neural or functional) state type with the following property: under optimal conditions, it is tokened if and only if p , and because p . What are optimal conditions? In the case of visual experience, we are in optimal conditions just in case we are in conditions of the sort that our visual system was designed to operate in by natural selection or by analogous processes in the course of ontogenic development. The same holds, *mutatis mutandis*, for other sorts of perceptual experience. This is still vague, of course; all extant proposals for the naturalization of intentionality are highly programmatic. Still, TTI is clear and definite enough for our purposes as it stands—clear enough, that is, to be challenged and defended.³

We will give the label “tracking representationalism” to the conjunction of representationalism with TTI. Tracking representationalism is, we think, the most promising view for the philosopher in search of a naturalistic account of experience. The purpose of this paper is to examine and respond to an apparently serious problem that pain experience raises for tracking representationalism. Very briefly, the problem, which will be laid out in more detail in section 2, is this: TTI seems to commit us to saying that the representational content of pain experience is exhausted by information about the location and physiological properties of some tissue damage or disturbance. But if tracking representationalism entails that this is all there is to the content of pain experience, then it is unable to account for one salient aspect of the phenomenology of pain, namely its negative affective quality. We offer arguments purporting to show that in order to account for this aspect of the phenomenology of pain, one must reject one of the conjuncts that define tracking representationalism—either representationalism or TTI. After developing this objection in section 2, we respond to it in section 3. In responding to the objection, we develop a picture on which our pain experiences do not just represent the presence of tissue damage, but also (roughly) represent our tissue damage as being bad for us to some degree. This view, we argue, is independently motivated by the phenomenology of pain experience, and we show how it is consistent with, and indeed predicted by, the tracking theory of intentionality. We conclude in section 4 with

some remarks concerning the advantages of our account of the content of pain experience over another account, which has recently gained some popularity among representationalists, according to which pain experience has imperative content.

2. Tissue Damage and the Painfulness of Pain

Consider your experience of a pain in a forearm. Under normal conditions, experiences of this type are caused by the presence of damage or disturbance of tissue in a forearm. Thus, it would seem that according to TTI, the content of your experience is something like: there is a disturbance of type d in location l .⁴ (Here we should assume that the physiological type d includes information about the shape, volume, and intensity of the disturbance, and perhaps also qualitative information about the disturbance, e.g. whether it is mechanically, thermally, or chemically induced.) In brief, tissue damage or disturbance is what pain experiences track, so it follows from TTI that this is what pain experiences represent. This view has the virtue of making sense of many aspects of the phenomenology of pain. For example, when we experience pain, the pain seems to be located in a particular bodily location, and it seems to have a distinctive shape and volume. When, as we say, someone has a stabbing pain in her arm, she experiences the pain as being in an arm, and she experiences it as occupying a fairly well-defined and bounded volume and as having a dagger-like shape. This is because her experience represents a disturbance with these properties, and it does so in virtue of the fact that her internal state stands in a tracking relation with these properties.

Many commentators, however, have noted that if tissue damage is all that pain experience represents, then it's hard to explain one salient aspect of its phenomenal character, namely its "negative affect."⁵ Pains don't merely inform us of the presence of some disturbance at a location in our bodies; they *hurt*. A theory which says that pain experiences merely represent tissue damage at some location in one's body would seem to leave out the painfulness of pain. This is not to say that pain experiences do not represent the physiological condition—the physical disturbance—of the relevant body parts. But that cannot be all there is to such experiences. For the physiological disturbance that you feel in your arm when you experience pain in your arm also feels *bad*.

According to some forms of representationalism, phenomenal properties of experiences, i.e., properties constitutive of what it's like to undergo those experiences, are identical to certain representational properties of those experiences, in particular properties which are constitutive of what the experience represents (as opposed to, say, intrinsic properties of the vehicles of representation), such as the property of ascribing redness to something,

the property of representing loudness, the property of having the content that *p*, etc. And most versions of representationalism at least accept the weaker claim that for every phenomenal property *P*, there is a representational property *R* such that, necessarily, an experience has *P* iff it has *R*. Take, for example, the phenomenal property—call it phenomenal reddishness—distinctive of experiences as of red things. Plausibly, the representational property corresponding to reddishness, i.e., the representational property *R* such that necessarily, an experience has phenomenal reddishness iff it has *R*, is the property of representing something as red. But now consider the property of pain experiences—call it painfulness—that we allude to when we say that a pain in the leg feels bad. What is the corresponding representational property in virtue of which our pain experiences are painful?

In light of the phenomenology of pain, it's natural for the representationalist to hold that our pain experiences not only represent the presence of a bodily disturbance, but also represent that disturbance as *bad*, or as having some such negative valuational property. The problem is that this position does not seem to be available to the tracking representationalist. On tracking representationalism, your experience only gets to represent that something has a property if you are in an internal state that is causally sensitive to exemplifications of that property. But it might be thought that the property of being bad is not a causally efficacious property. Therefore, it is not a property to which your internal states can be causally sensitive. Murat Aydede (2006), in response to a paper⁶ in which one of current authors proposes that pain experiences represent the presence of tissue damage and represent that tissue damage as bad, puts this objection forcefully:

[Valuational properties like goodness and badness] are simply not the kind of qualities that can be detected or tracked (in the technical information-theoretic sense). One reason for thinking this is [...] there could be a molecule-by-molecule identical tissue damage which is not bad (not just experienced as bad, but just not bad—if we abstract away from the regular connotations of 'damage' and think of it as composed of whatever physical features and their configurations constitute the damage). There doesn't seem to be any natural property of tissue damage simple and suitable enough to allow itself to be transduced. (p. 131)

Note that the worry here is not that badness is not instantiated, that no thing (state, event, . . .) is bad. The objector can allow that some things really are bad, and indeed that the tissue damage that our pain experiences track is bad. But it is not enough that the tissue damage which experiences of pain track is bad. According to TTI, in order to represent the damage as bad, the badness must (under optimal conditions) be causally relevant to the person's being in the internal state that she is in. Compare: In having a

visual experience as of a green apple, you visually represent the color green. It happens that green is Smith's favorite color, but it does not follow (and it is not the case that) you visually represent that there is a surface before you having Smith's favorite color. This is because your internal perceptual state is causally sensitive to the colors of external objects, but it is *not* causally sensitive to whether a color is favored by Smith. Likewise, when you undergo a pain experience, your internal state is causally sensitive to many physiological features of the bodily disturbance, such as its location, shape, and volume, but according to the objection under consideration, your internal state is not causally sensitive to valuational properties of the disturbance, such as its badness.

Let us try to distill these considerations into a succinct and explicit argument. To begin with, the opponent of tracking representationalism cites empirical facts about what is tracked by the internal states we undergo during pain experiences to argue that tracking representationalism is committed to the claim that our pain experiences represent bodily damage or disturbance. And on the assumption that valuational properties like badness cannot be tracked by our internal states, she argues that tracking representationalism should hold that (the physiological type and location of) a bodily disturbance is all that pain experiences represent. Or in any case, if there is anything else represented by our pain experiences, it is hard to see how it could be anything relevant to explaining the painfulness of pain. These considerations are taken to support the following premise:

- (1) If TTI is true, then the content of pain experience is exhausted by information concerning the location and physiological type of a bodily disturbance.

Next, the objector claims:

- (2) If the content of pain experience is exhausted by information concerning the location and physiological type of a bodily disturbance, then a certain phenomenal feature of pain experience—the “painfulness of pain”—is not explained by its content.

From here, the objector makes the valid inference to:

- (3) Either the content of pain experience is not exhausted by information concerning the location and physiological type of a bodily disturbance, or the phenomenal features of our experiences cannot in general be explained by their content.

From here, one can almost conclude that tracking representationalism is false. For from the first disjunct of (3) together with (1), we can infer that

TTI is not true by *modus tollens*. Since tracking representationalism is just the conjunction of TTI with representationalism, it follows that tracking representationalism is false. And the latter disjunct of (3) seems simply to deny the truth of representationalism, and hence entail the falsity of tracking representationalism. In either case, tracking representationalism is false.

However, as we are defending representationalism in the present context, the representationalist is only committed to the supervenience of phenomenal character on representational content, which in itself does not entail that the phenomenal character of an experience can always be explained by its content. (Perhaps there is some sort of “explanatory gap” between the representational facts about an experience and the phenomenal facts about an experience even though the former facts necessitate the latter.) Even so, in our view the tracking representationalist ought not respond to the above argument by citing such an explanatory gap. If there is an explanatory gap that is unique to phenomenal consciousness, it has to do with our inability to understand why, given that a state has such-and-such representational properties, it should have any phenomenal character *at all*. We do not think that there is any very interesting explanatory gap having to do with an inability to understand or explain why, given that a state has such-and-such representational properties, it has *this* phenomenal character rather than some other phenomenal character. (If there is an explanatory gap here, it is the same sort of “explanatory gap” involved in understanding why there is water rather than mercury around given that there is H_2O around, or understanding why this apple is red rather than blue given that it has such-and-such a spectral reflectance.) The tracking representationalist who cited an explanatory gap to respond to the above argument would have to cite the latter sort of explanatory gap. The consequent of (2) says that a certain phenomenal feature of our pain experiences—the painfulness of pain—is not explained by their content. What’s not explained, according to the intended reading of the consequent of (2), is why pain experiences should be painful rather than having a phenomenal character that does not involve painfulness.

Moreover, if the tracking representationalist concedes that the content of pain experience cannot explain this aspect of its phenomenal character, this undermines our reason to accept that the latter supervenes on the former. Lacking an explanation of why, given that a pain experience represents the presence of a bodily disturbance, it must have the phenomenal property of being painful, we have at least some reason to think that it is possible for an experience to have the former (representational) property without having the latter (phenomenal) property. It seems possible, after all, that a purely tactile experience could represent that there is some physiological disturbance at a certain location in our bodies without that experience having the negative affect involved in typical pain experiences.

The view that it is indeed possible for two experiences to differ with respect to whether, or the degree to which, they are painful without differing with respect to what sort of bodily disturbance they represent has been argued for recently by Adam Pautz (2010). Consider the following scenario:

Mild and Severe: Two communities of pain-perceivers evolve separately. Mild belongs to one community and Severe belongs to the other community. Both occasionally experience bodily disturbance d in the leg. In Mild's community, d is not very dangerous. So d normally puts his primary somatosensory cortex into state m involving a certain mild rate of firing of neurons. [. . .] By contrast, in Severe's community, d is much more dangerous. For instance, maybe it is more susceptible to dangerous infections in this community because the community occupies an environment in which bacteria are more plentiful. In consequence, in Severe, d normally causes somatosensory state s , involving a rate of firing of somatosensory neurons which is twice as great as that which is involved in m . As a result, Severe attends to his leg with greater urgency than does Mild. But optimal conditions obtain in each case, because the different behaviors are completely adaptive given the noted difference in the significance of the damage to them. So, Mild and Severe bear the optimal cause relation [i.e., what we have been calling the "tracking" relation] to the same property, d , but they differ radically in pain processing and behavior. (p. 34)

It would seem that TTI predicts that the pain experiences of Mild and Severe have the same content. Both of their experiences represent that there is some tissue damage of a certain specific sort at a certain bodily location. This, together with representationalism, yields the result that Mild and Severe have phenomenally identical pain experiences—i.e., their pain experiences are exactly alike with respect to phenomenal character. But this prediction looks wrong. We can imagine that when Severe undergoes d , he cries out in apparent agony, attending with great urgency to his wound, whereas when Mild undergoes d , he behaves as though he is only very slightly perturbed. It would be odd to hold that, nonetheless, their pain experiences have the same phenomenal character. Surely Severe is experiencing a very severe pain, while Mild is experiencing a very mild pain.⁷

A tracking representationalist might reply that that the difference between Mild and Severe lies in the strength of their desires to rid themselves of d , not in the phenomenal character of their pain experiences. This seems implausible. When people desire (and try) to rid themselves of pain, it is *because* the pain feels bad. Of course there will be a difference in the strength of their desires to rid themselves of d , but this difference should be explained by the difference in how bad their pains feel. Severe desires to rid himself of d much more strongly than Mild does because Severe's pain feels much worse than Mild's pain does. We should say that in this scenario, Mild and Severe have experiences that differ in phenomenal character, even though it seems to be a consequence of TTI that their experiences have the same content.

And since the scenario of Mild and Severe is clearly a possible scenario, it looks like we can conclude that:

- (4) If TTI is true, then possibly, there are experiences e_1 and e_2 such that e_1 and e_2 differ in phenomenal character but do not differ in representational content.

The consequent of (4) is equivalent to the negation of:

- (5) Necessarily, for any experiences e_1 and e_2 , if e_1 and e_2 differ in phenomenal character, then they differ in representational content.

And (5) is just the thesis of representationalism (as we are considering it here). So, the material conditional (4) is equivalent to the negation of the conjunction of TTI with representationalism—that is, (4) is equivalent to the denial of tracking representationalism.

3. Tracking Representationalism Defended

How should the tracking representationalist respond to the challenge presented in the previous section? The arguments given above each hinged on the assumption that the content of a pain experience is exhausted by information concerning the type and location of some bodily disturbance. We saw earlier that in order to account for the “painfulness of pain,” the representationalist is pressured to say that pain experiences, in addition to representing the location and physiological type of a bodily disturbance, represent that disturbance as bad, or as having some such negative valuational property. But it is natural to think that such valuational properties are epiphenomenal; they are not causally efficacious in the way that is required for us to represent them on a tracking theory of representation.

Our response to the challenge presented in the previous section will be to argue, roughly, that in fact the valuational properties like badness are causally efficacious; their exemplifications influence what happens in the world, and moreover, such properties are tracked by our internal states in the relevant way. The plan for this section is as follows. First, we clean up the picture of what the representationalist should say about the representational content of pain experience in light of its phenomenology. What will emerge is a view on which the pain experience of an individual A represents that there is a bodily disturbance of a certain type at a certain bodily location, and that this bodily disturbance is bad for A to a certain degree. Next, we give a naturalistic account of what the property of being bad for A is. Finally, we argue that this property is causally efficacious, and moreover that

it is a property which is tracked by the internal states we undergo when we experience pain.

Consider again Pautz's scenario of Mild and Severe. Intuitively, the difference between Severe's experience and Mild's experience consists in the fact that when Severe undergoes bodily disturbance d , it feels much worse to him than it feels to Mild when he undergoes the same bodily disturbance. This suggests that we do not merely represent our bodily disturbances as bad in undergoing pain experiences; we also represent them as bad to some specific degree. Badness is a gradable determinable property. Its determinates are specific degrees of badness, and like other gradable determinable properties, our experiences can represent its various determinates. Consider brightness, loudness, or sweetness. Visual experience doesn't merely represent a stimulus as bright. It represents it as having some specific degree of brightness, such that while looking at two bright objects, one object can look brighter than the other. Similarly, auditory experience represents stimuli as being loud to some specific degree, so that some sounds can be experienced as louder than others; and taste experience can represent food as being sweet to some specific degree, so there can be two sweet things, one of which tastes sweeter than the other. Similarly, an adequate account of the content of pain experience must understand pain experiences as capable of representing a range of specific degrees of badness.

Further, we propose that the representationalist should hold that an individual's pain experience represents some bodily disturbance as not simply bad (to a certain degree), but rather as bad *for her* (to a certain degree). That is, the valuational property represented in pain experience is subject-indexed. Subject-indexed properties are ubiquitous in the contents of experience. For instance, my visual experiences do not represent the absolute locations of objects. They represent the subject-relative locations of objects—the distance of various objects away from me, whether they are in front of, to the left of or to the right of me, and so forth. The same holds for spatial representation in auditory experience.

Why think that the valuational content of pain experience involves subject-indexed properties in this way? One reason will emerge later; we will argue that the valuational property that features in the content of pain experience should be identified with a certain naturalistically acceptable property, and it happens that this property is more naturally identified with a subject-indexed valuational property. But there are other independent reasons to accept that the relevant property is subject-indexed. For suppose that pain experiences instead just ascribed the property of being bad to a certain degree, rather than ascribing the property of being bad for the subject to a certain degree. Then Mild's experience represents the bodily disturbance d as being bad to degree x , while Severe's experience represents d as being bad to some greater degree y . Being bad to degree x and being bad to degree y are two determinate properties of the determinable badness. Typically, whenever

there are two distinct absolute determinates of a single determinable, the two determinate properties are contrary properties in the sense that they cannot be co-instantiated. Since being-bad-to-degree- x and being-bad-to-degree- y are determinates of a single determinable property badness, and neither is a determinate of the other, the disturbance quality d could not have both of these properties. But then it would follow that either Mild's experience or Severe's experience would be falsidical. But intuitively, there is nothing falsidical in either of their experiences.⁸ We can allow that both of their experiences are veridical if we allow the relevant negative valuational property to be subject-indexed. There is nothing inconsistent about the co-instantiation of being bad for Mild to degree x and being bad for Severe to degree y , any more than it is inconsistent to have a single thing co-instantiate being five feet away from me and being ten feet away from you.

We have argued for a somewhat more refined account of what the representationalist ought to say about the content of pain experiences to account for the painfulness of pain. The content of a pain experience of an individual A is something like: there is a bodily disturbance of (physiological) type d in location l , and d is bad for A to degree x . But the problem for tracking representationalism remains. If TTI is correct, then in order for A 's experiences to represent a bodily disturbance as being bad for her to a certain degree, A 's internal states must be causally sensitive to such valuational properties. But how can A 's internal states be causally sensitive to the degree to which something is bad for her?

To answer this, we must first ask: what is the property of being bad for a given individual? In your own case, patently some things are bad for you, e.g., undergoing tissue damage, eating poisonous mushrooms, and inhaling large quantities of smoke. Assuming a naturalistic worldview, then, we should not suppose that being-bad-for-you is a spooky or primitive non-natural property. Plausibly, the property of being bad for you is just the property of being apt to harm you. Indeed, the following modal equivalence claim seems to hold, at least on a certain sense of "bad for you."

(6) Necessarily, x is bad for you iff x is apt to harm you.

If (6) is correct, then the properties *being bad for you* and *being apt to harm you* are necessarily co-instantiated. On some views, this entails their identity. More cautiously, we can say that it at least gives us good reason to identify them. If what it is for something to be bad for you is to be apt to harm you, what is it for something to harm you? While the notion is difficult to make precise, we take it that there is no great mystery about what it is for something to harm something else, or about how it is consistent with a naturalistic worldview that something could be harmed. Since Darwin, we are all now familiar with how the notion of a teleological system can be made naturalistically acceptable. We can understand the notion of harm

in relation to the notion of a teleological system. *Very* roughly, something harms a teleological system to the extent that it hinders that system (or one of its subsystems) from performing its function(s).

So if the property of being bad for you is the property of being apt to harm you, can your internal states be causally sensitive to it in such a way that they represent this property? They can. To see why, let us return to the formulation of TTI. According to TTI, tokens of a state type S in an individual x represent that p if and only if:

- (7) Under optimal conditions, x tokens S iff p and because p .

Since (7) can only be satisfied by values of S which are the kinds of entities that can be tokened, values of S must be state types, but it is left open whether the state types that satisfy (7) will be physically or functionally defined. Now consider again the scenario of Mild and Severe. On our view, Mild's experience represents that (there is a disturbance d and d is bad for Mild to (some fairly low) degree x), whereas Severe's experience represents that (there is a disturbance d and d is bad for Severe to (some fairly high) degree y). According to TTI, then, Mild must be tokening a state S such that

- (8) Under optimal conditions, Mild tokens S iff there is a disturbance d and d is bad for Mild to degree x and because this is the case.

and Severe must be tokening a state S' such that:

- (9) Under optimal conditions, Severe tokens S' iff there is a disturbance d and d is bad for Severe to degree x , and because this is the case.

We suggest that the value of S satisfying (8) is a functional state type F defined (very roughly) by its forward-looking causal role of bringing about "mild" avoidance behavior.⁹ And the value of S' satisfying (9) is a functional state type F' defined (very roughly) by its forward-looking causal role of bringing about "severe" avoidance behavior. Are tokens of F (F') in Mild (Severe) caused by the fact that there is a disturbance d and d is bad for Mild (Severe) to degree x (y) under optimal conditions? Quite plausibly, yes. For plausibly, the explanation for why Mild tokens F —a functional state defined by its role in producing "mild" avoidance behavior—when he undergoes d is that d is, to a small degree, apt to harm him. Likewise, the explanation for why Severe tokens F' —a functional state defined by its role in producing "severe" avoidance behavior—when he undergoes d is that d is, to a very large degree, apt to harm him. The explanations in question are causal explanations. Members of Severe's species were designed by natural selection to token states that bring about "severe" avoidance behavior when they undergo disturbance d because d is highly dangerous for members of

Severe's species—that is, because it is very much apt to harm them. The same goes, *mutatis mutandis*, for members of Mild's species.¹⁰

Perhaps one is comfortable with saying that, for example, when Severe undergoes *d*, Severe tokens *F* because *d* is highly apt to harm him, but uncomfortable with saying that *d*'s being highly apt to harm him *causes* Severe to token *F*. We agree that not all true sentences of the form “*a* is *F* because *b* is *G*” imply sentences of the form “*b*'s being *G* causes *a* to be *F*”. Sometimes sentences of the former sort are made true by some other, non-causal dependence relation holding between *a*'s being *F* and *b*'s being *G*. (Consider, for example, the sentence “The proposition that snow is white is true because snow is white.” This doesn't imply “snow's being white causes the proposition that snow is white to be true.”) In the case at hand, however, we don't think that there is any plausible alternative dependence relation that could account for the truth of the relevant “because” claim. Perhaps the suspicion that the property of being bad for one cannot be causally relevant to one's tokening a certain functional state arises from something like the following line of thought: The property of being bad for one does not supervene on the local physical properties of a bodily disturbance. Whether a disturbance is apt to harm an organism depends also on facts about the organism as a whole together with facts about the environment in which the organism is embedded. But a property can only be causally relevant in producing an effect if the property supervenes on local physical properties. So the property of being bad for one cannot be causally relevant in bringing about the tokening of a certain functional state.¹¹

The problem with this argument is that it's just not true that a property can only be causally relevant in producing an effect if the property supervenes on local physical properties. When John tells Jane that she looks like a cow, Jane slaps him. John's utterance causes Jane to slap him, and does so in virtue of its semantic properties. But the semantic properties of an utterance do not supervene on its local physical properties; rather, they depend (perhaps *inter alia*) on facts about how words are used in one's community. Mental properties, like having the desire to drink water, are causally relevant in producing behavior. But if we accept the standard arguments for mental externalism, (many) mental properties, including the property of having a desire to drink water, do not supervene on local physical properties. But it would be silly to say that my having a desire for water has never caused me to drink water. Or consider properties, like the property of being a lion, which depend not just on an object's local physical properties but also on its historical origins. A herd of gazelles runs away upon encountering a lion. Surely the fact that the object they encountered was a lion causally explains (at least in part) their fleeing the scene. These are truisms about causation. There are theories of causation that entail that the only causally relevant properties are those which supervene on local physical properties, and therefore have the consequence that the cases listed above are not

genuine cases of causation. On our view, a constraint on any adequate theory of causation is that it accommodate the truisms discussed above. That some theories of causation have the consequence that these cases are not genuine cases of causation should be taken as a *reductio* of those theories.

In any case, the point is not crucial. The official formulation of TTI only requires the truth of the relevant “because” claim. It is neutral on whether the dependence relation that makes it true is a causal one.¹²

There are two sorts of cases that might seem to make trouble for our proposal. As a way of clarifying our position, we conclude this section by addressing these cases and showing why they are not as problematic as they may appear. The first sort of case involves bodily disturbances which feel bad (and so, on our view, are represented as being bad for us), but which in fact are good for us in the circumstances. Such cases are familiar. The nurse punctures my arm with a needle in order to inject a vaccine. A man has his leg amputated (without anesthetic) to prevent a tumor from spreading to the rest of his body. In these cases, the resultant experience represents the bodily damage in question as bad for one, when in fact it is not bad for one in the circumstances. These cases are not a problem for our view, since under optimal conditions (roughly, the conditions we were designed to operate in by natural selection), getting one’s leg cut off or having one’s skin punctured is bad for one. It is because of this fact that, when we undergo such tissue damage, we token functional states characterized by their forward-looking functional role of bringing about aversive behavior. And this, on our theory, is what makes it the case that these token states represent the disturbance in question as bad. None of this, however, requires that the disturbance is bad for one in one’s actual present conditions.

The second sort of case involves bodily disturbances where the degree of felt badness does not seem to correspond with the degree to which it is apt to harm one, even in design conditions. These are well illustrated by looking at pairs of bodily disturbances which vary considerably with respect to how harmful they are (in design conditions) but do not vary much with respect to their felt badness. Case 1: As a result of an accident, your leg is sliced open at the location of your great saphenous vein. Case 2: As a result of an accident, your leg is sliced open just next to the great saphenous vein. In the first case, the tissue damage is extremely harmful. In the second case, the damage is somewhat harmful, but much less so than in the first case. However, the felt badness of the pain is likely to be approximately the same in each case. Let d_1 and d_2 be the disturbance types represented, respectively, by your experiences in cases 1 and 2. Let’s suppose also that the two experiences are identical with respect to felt badness, so each experience represents the relevant damage as being bad for you to the same (moderate) degree x . Let’s also suppose that d_2 is in fact bad for you to degree x under design conditions. On our proposal, your experience in case 1 represents d_1 as being bad for you to degree x , and

does so (partly) in virtue of the fact that it is a token of a functional state F such that, under optimal conditions, F is tokened because d_1 is bad for me to degree x . But how can this be, since the disturbance you undergo in case 1 is bad for you (even under optimal conditions) to a degree much higher than x ?

To understand what's going on here, we must take note of the fact that your current token bodily disturbance falls under multiple distinct types. For instance, it is a disturbance of the type *slice of the great saphenous vein*; this type of damage is bad for you (under optimal conditions) to a very high degree (a degree much higher than x). But it is also a disturbance of the types *slice of the leg*, and *slice of the skin*. These types of damage are bad for you to a much lesser degree. If your experience in case 1 represents your disturbance under the type *slice of the great saphenous vein*, then this would be a problem for our view. But just because your (token) disturbance does belong to this type, it does not follow that your experience represents it under this type. It is quite plausible that the type under which your experience represents the disturbance in case 1 (namely d_1) is very similar to, or even identical to, the type under which your experience represents the disturbance in case 2 (d_2). (Perhaps they are both represented under a type like *slice of the skin*.) If so, then it's plausible to suppose that d_1 and d_2 in fact are equally bad for you under optimal conditions.

4. Concluding Remarks: Against Imperative Content

Recently, a number of representationalists have invoked the notion of imperative content to account for the painfulness of pain.¹³ Roughly, the idea is that we account for the sensory dimension of pain experience in terms of the familiar indicative contents concerning the location and physiological type of some bodily disturbance, and we account for the negative affective dimension of pain in terms of imperative contents. So in addition to having the indicative content that there is a disturbance of type d at location l , pain experiences also have an imperative content like: stop that bodily disturbance! Such accounts are often motivated by the difficulty in accounting for the negative affective dimension of pain by appeal to its indicative content. They thus assume that the indicative content of pain experience must be exhausted by information concerning the location and physiological type of a bodily disturbance. Inasmuch as this assumption stems from the thought that the most promising naturalistic theories of intentionality cannot deliver valuational contents of the sort that could account for the painfulness of pain, we hope that the previous section has undermined these motivations for accepting an imperative content account of pain. We conclude this paper by presenting two reasons why the representationalist should prefer our account to the imperative content account.

First, as we have noted, the felt badness of pain comes in degrees. Any adequate account of the painfulness of pain must not only account for the fact that pain is painful, but also for the fact that pain can be painful to different degrees. Our account does so by proposing that our pain experiences ascribe determinates of the gradable determinable property *badness for oneself* to bodily disturbances. But how can the proponent of imperative content account for this? Consider again Mild and Severe. Both have an experience as of a bodily disturbance of the same physiological type in the same location, and both undergo an experience with a negative affect. A proponent of imperative content should hold that the indicative contents of their experiences are the same, as both experiences should have as their indicative content that there is a disturbance *d* at such-and-such location. They will then account for the negative affect of their pain experiences by saying that their experiences have, in addition, the imperative content: stop that bodily disturbance! But how, then, can they account for the difference in their experiences? For the representationalist, any difference in phenomenal character between two experiences must be accompanied by a difference in content, but it doesn't look as if the proponent of imperative content has the resources to deliver a difference in content between Mild's experience and Severe's experience.

Perhaps the proponent of imperative content could say that states with the same imperative (and indicative) content can nonetheless differ along some other dimension—which we might call the urgency of the state—which is also relevant to determining phenomenal character. On such a proposal, the difference between Mild's experience and Severe's experience is analogous to the difference between the following two imperative sentences:

- (10) (Please) stop that bodily disturbance.
- (11) Stop that bodily disturbance!!!

But this approach is still objectionable.

First of all, imperative contents are usually invoked by representationalists in order to account for the painfulness of pain in terms of the content of pain experience. Unfortunately, this more complicated imperative content account is inconsistent with representationalism. For now the phenomenal character of an experience does not supervene on its content alone; rather it supervenes on its content together with its degree of urgency. More importantly, this approach adds needless complexity and inelegance to a representationalist account of phenomenology. Our account has the advantage that its treatment of the various degrees of the felt badness of pain is structurally similar to the most plausible treatments of, e.g., the various degrees of apparent brightness, loudness, and sweetness in experience. The most plausible account of the latter has it that experience represents determinates of the gradable determinable properties brightness, loudness,

and sweetness. We see no good reason to give a fundamentally different account of variation in pain intensity. To do so is to adopt a complex and inelegant theory when a simpler and more elegant theory is available.

A final reason to prefer our account over the imperative content account is that our account generalizes to other “affectively valenced” experiences far more naturally than the imperative content account. We have focused on the painfulness or negative affective quality of pain experience. But pain is just one of a number of affectively valenced experiences. There are other experiences with a negative affect, such as the experience of smelling excrement or eating rotting meat. More importantly for the present discussion, there are also experiences, such as the experience of having an orgasm or eating a ripe strawberry, which have a positive affect. We say that strawberries taste good and orgasms feel good, and intuitively, when we say such things we are (at least obliquely) commenting on the phenomenal character of our experience of strawberries and orgasms. Our account of the content of pain experiences has a natural extension to affectively positive experiences: when an individual *A* has such an experience, the experience represents its object as good for *A* to a certain degree. The property of being good for *A* will be identified with the property of being apt to benefit *A*, and the explanation for how our experiences come to represent this property will proceed along the same lines as the explanation given above for how pain experience comes to represent the property of being bad for one. But how can the proponent of imperative content extend her proposal to account for the positive affect of some experiences?

Consider the experience one has when one chews on a delicious strawberry. On the imperative content view, presumably one’s experience (in addition to whatever indicative content it might have) has an imperative content like: continue eating that strawberry! (Eat more of that kind of thing?) But what is it about this content that accounts for the positive affect of one’s experience? It cannot be merely the fact that it is an imperative content, for the imperative content theorist holds that pain experiences, which are affectively negative, also have imperative content. So there must be some sort of relevant difference between the imperative content of pain experience (stop that damage!) and that of the experience of eating a strawberry (continue eating that strawberry!) that accounts for the fact that the former experience is affectively negative while the latter is affectively positive. Of course there are differences, but it is not at all transparent why any of these differences should be relevant to accounting for the difference in valence between the two experiences. By contrast, on our account, there is a very straightforward explanation for why our experience of pain has a negative affect and our experience of eating a strawberry has a positive affect. Pain feels bad to us and strawberries taste good to us because our experiences of pain represent their objects as bad for us and our experiences of strawberries represent their objects as good for us.

Notes

1. This thesis is sometimes called ‘weak representationalism’. Strong representationalism is the thesis that the phenomenal character of an experience is one and the same as its representational content. For more on different forms of representationalism, see Tye (forthcoming).
2. This does not entail that it is the neurophysiological properties of the state that make that state an experience. Compare: a token sunburn is identical with a particular physical state of one’s skin, but it is not (merely) the intrinsic physical properties of the state of one’s skin that make that state a sunburn. One might hold that having representational content is an essential property of experiences and that states have content not in virtue of their intrinsic physical properties, but in virtue of their relations to external objects and properties, in which case one may accept that token experiences are identical with token neural states while denying that it is the neurophysiological properties of the state which make that state an experience. One further comment: sometimes it is held by naturalists that token experiences are constituted by brain states without being identical with them. This is our own preferred view but for ease of exposition we ignore it in what follows.
3. Aside from the vagueness of “optimal conditions,” TTI faces obvious problems in accounting for the contents of many of our cognitive states, e.g., the belief that Zeus lives on Mount Olympus or the belief that there is no greatest prime number. The fact that there is no greatest prime number is not causally efficacious, so my internal states cannot be causally sensitive to this fact in the way that TTI requires in order for my internal states to have the content that there is no greatest prime number. TTI is most promising in accounting for the content of perceptual states, and since this paper is concerned with perceptual states rather than cognitive states, we will not focus on the peculiar problems raised by cognitive states for TTI. Quite plausibly, a different theory is required to account for the contents of cognitive states than is required to account for the contents of perceptual states. Arguably, belief states are importantly different from perceptual states in that belief states, unlike perceptual states, involve sentential representations whose contents are determined compositionally—that is, their contents are fixed by the contents of their simpler constituent representations together with their syntactic organization. If so, then it is plausible that a theory of belief content, unlike a theory of perceptual content, will explain how our beliefs have the propositional content they do by first explaining how the simpler cognitive representations involved in belief acquire their content and how syntactic relations among cognitive representations come to encode semantic projection rules. Since we will be concerned with TTI only insofar as it offers a theory of the contents of perceptual states, we will ignore the problems mentioned above with applying TTI to cognitive states.
4. Whether the content is really existential (as suggested in the text) or singular is a complicated issue. The corresponding issue arises for perceptual experiences. For a detailed discussion of the singular/general debate with respect to perceptual experience, see Tye (2009) and (forthcoming).
5. See, e.g., Aydede (2001).

6. Tye (2006).
7. Pautz gives further neurological evidence that the pains that Mild and Severe experience would differ in intensity. He notes that, in humans, there is a linear relationship between firing rates in primary somatosensory cortex and the reported intensity of pain experience. Since we are told that Severe's somatosensory state involves a firing rate twice as great as that involved in Mild's somatosensory state, there is some further reason to think that Mild and Severe would experience pains of different intensities.
8. We have been assuming that pain experience ascribes the relevant negative valuational property to a shareable quality that can be exemplified by bodily regions. Instead one might think that, for example, Mild's and Severe's pain experiences ascribe the relevant valuational property, not to the shared bodily disturbance quality *d*, but to token events/states of exemplifying *d*—that is, not to *d* but to token exemplifications of it. The above argument does not threaten such a view, since there is no inconsistency in saying that contrary properties are possessed by two distinct token events/states. (Exemplifications of a property or quality should not be confused with tropes of that property or quality, though a view on which the relevant valuational property is ascribed to tropes of bodily disturbance qualities will be similarly unthreatened by this argument.) But even supposing such a view is right, a more *recherché* scenario can be used to show that the relevant valuational property is subject-indexed. For consider a scenario in which surgeons connect Mild and Severe together at the hip, so that they both feel pain when there is damage to the bodily region at which they are connected. When *d* is exemplified at that region, each undergoes the same token state of bodily damage. If, rather than ascribing subject-indexed valuational properties, Severe's experience ascribes badness-to-degree-*y* to this token state and Mild's experience ascribes badness-to-degree-*x* to it, then at least one of their experiences would have to be falsidical. But again, there would seem to be nothing falsidical about either of their experiences.
9. One reason that we take *S* to be a functional state type is that there could be two neural state types, N_1 and N_2 , each of which is tokened in Mild under optimal conditions only if (and because) there is a disturbance *d* in Mild's body that is bad for Mild. In these circumstances, neither N_1 nor N_2 represents that there is a disturbance *d* that is bad for Mild, since the 'if' condition in the 'iff' clause required by TTI is not met. So, either *S* is a disjunctive state (N_1 or N_2) or it is a functional state realized by N_1 in some cases and by N_2 in others. We prefer the latter alternative. As noted below in the text, this alternative, if it is cashed out further in terms of the forward-looking functional state proposed in the text, has the further virtue of offering a straightforward explanation of why *S* is tokened in Mild under optimal conditions iff and because there is a disturbance *d* that is bad for Mild; for intuitively it is because Mild is subject to a disturbance that is apt to harm him that his state causes him to engage in avoidance behavior. Finally, taking *S* to be a forward-looking functional state provides an overall view of pain that is systematically unified. What it is for an individual to be in pain is for the individual to token a forward-looking functional state with a backward-looking aspect that gives the state a certain representational content, which content 'fixes' the phenomenal character of pain (its painfulness).

10. One might be tempted to say that the explanations in question are teleological rather than causal. But the explanations are not teleological explanations in any sense of “teleological” such that teleological explanations and causal explanations are mutually exclusive. There may be such teleological explanations, e.g. “I bought the tickets in order to go to the show,” or “My heart is pumping blood in order to deliver oxygen to various parts of my body.” In these cases, I do not cite in the explanans any event or fact that obtains prior to the explanandum, so perhaps such explanations are not causal. However, the explanations at issue clearly do not have this form. If there is any interesting sense in which the explanations in question are teleological, it does not seem to be a sense in which teleological explanations and causal explanations are mutually exclusive.
11. Something akin to this line of thought is at play in the passage from Aydede (2006) quoted in section 2.
12. There are close cousins to TTI which do not explicitly invoke any sort of dependence relations. For example, Dretske’s (1988) theory of representation invokes the notion of indication. Very roughly, tokens a state type *T* indicate that *p* iff: if *T* is tokened, *p*. In many cases, indication relations will hold in virtue of causal or other dependence relations, but this is not required by the definition of indication. Dretske incorporates the notion of indication into an account of representation roughly as follows: tokens of a state *S* represent that *p* just in case it is the biological function of *S* to indicate *p*. It is overwhelmingly plausible that the internal states we undergo when we experience pain have the biological function of indicating that some bodily disturbance is apt to harm us. And on the Dretskean account, questions about exactly what sort of dependence relation holds between our internal states and the aptness of some bodily disturbance to harm us do not arise. We take Dretske’s theory to be broadly in the same spirit as TTI, and we do consider it a competitor for the purposes of this paper.
13. See, e.g., Martinez (2011) and Klein (2007). Hall (2008) proposes that other bodily sensations like the feeling of an itch or the feeling of hunger have imperative contents. The first of our criticisms of the imperative content account of pain below can be very naturally extended to an imperative content view of these other bodily sensations.

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